AKILAN P L

B.E. CSE (AI/ML) Student

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PROFILE

AI/ML engineering student specialising in production RAG systems and on-premises LLM deployment, with end-to-end experience building enterprise AI platforms at KONE. Strong Python and backend foundation; SIH 2024 Finalist; open-source contributor to PyTorch. Targeting Software Engineering / ML Engineering internships.

PROFESSIONAL EXPERIENCE

Artificial Intelligence Intern (Enterprise Innovation Team), ***KONE*** **12/2025 – 01/2026**

Chennai, Tamil Nadu, India

- Built an on-premises RAG platform using Python, LangChain, and FAISS, enabling secure conversational access to 100+ enterprise documents — achieving ~30% lower query latency and ~40% improvement in retrieval relevance vs. baseline keyword search.
- Automated PDF ingestion & extraction for 100+ enterprise documents using Python (PyMuPDF, pdfplumber), reducing manual processing time by an estimated 80% and eliminating copy-paste errors across reporting workflows.

Campus Ambassador, ***Perplexity.ai*** **09/2025 – 11/2025**

Chennai, Tamil Nadu, India

- Ran hands-on AI tool workshops for 100+ students, driving adoption of AI-assisted research workflows on campus.

TECHNICAL CONTRIBUTIONS

PyTorch (Meta) Open Source Contribution — Serialization Security Documentation, ***github.com/pytorch/pytorch/pull/172743***

- Identified and documented a serialization security risk in `torch.export` (`torch.load` with `weights_only=False`); proposed safer loading paths using `weights_only=True` and safetensors.
- Contribution merged into the PyTorch core repo, strengthening security guidance for millions of developers.

ACHIEVEMENTS

Winner — Hexaware GenAI Designathon 2026 | ***Team Lead*** **|** ***Lead Developer***

- Led a 2-member team to build an intelligent survey data-quality platform combining statistical analysis, machine learning, behavioural patterns, and historical trends to detect anomalies beyond conventional validation.
- Unified multiple signals into explainable risk scoring, enabling supervisors to prioritize investigations and accelerate data-quality review across large survey datasets.

PROJECTS

Enterprise RAG Platform, ***github.com/akilanpl/enterprise-rag-platform*** **11/2025 – 01/2026**

- Developed an on-prem RAG system for enterprise knowledge access, achieving ~30% lower query latency, ~40% improvement in retrieval relevance, and supporting ingestion/search across 100+ internal documents with zero external data exposure.

Offline Mapping & Navigation System (SIH 2024 FINALIST) **08/2024 – 04/2025**

- Built an offline geospatial navigation pipeline using Python to detect highway vs. service-road movement through tile maps, heading vectors, and coordinate clustering algorithms.

Climate Detection Using AI **08/2024 – 12/2024**

- Built a CNN-based climate classification model with custom preprocessing and evaluation pipelines; published research.

SKILLS

Core Languages
Python (primary), SQL, JavaScript, Java, C

AI/ML
RAG, LangChain, FAISS, on-prem LLM deployment, document ingestion & retrieval, CNN, model evaluation, secure model loading (safetensors)

Cloud
AWS (EC2, S3, IAM, networking), Azure (data workflows, ML), Google AI Studio

Backend & Systems
Flask, REST APIs, Python scripting, API development, automation workflows, Git / GitHub

Data & ML
Data pipelines, feature engineering, geospatial data processing (OSMnx, tile maps), model experimentation

EDUCATION

B.E. CSE (AI & ML), ***Rajalakshmi Institute of Technology*** **09/2023 – Present**

Chennai, Tamil Nadu, India

- Finalist, Smart India Hackathon 2024 — Offline Mapping & Navigation System; Co-author of 2 academic research papers in AI/ML — Published at IJITRA.

CERTIFICATES

AI Agents Intensive Program - Google

AWS Cloud Foundations - AWS Academy

Azure for Data Science - Microsoft